

UNIT-4

Ques

Q1. Outline the Diffie-Hellman Key Agreement process.

Ans. The Diffie-Hellman Key Agreement (or key exchange) is a method for two parties to securely share a cryptographic key over a public channel.

Working:-

① Initialize with public parameters

- Both parties agree on two public parameters

(a) Prime no. p - a large prime no.

(b) Generator g - a no. less than p with special properties in modular arithmetic. Typically a primitive root modulo p .

→ The parameters ' p ' and ' g ' do not need to be secret, so they can be shared over a public channel.

② Compute Public Keys

- Each party computes their public key

Ans) Purpose: To securely exchange cryptographic keys over a public channel.

• Concept: Both parties generate public and private values and exchange public values. Using each other's public values and their own private values, they compute a shared secret.

• Foundation: Based on the mathematical difficulty of solving the Discrete logarithm problem in modular arithmetic.

if $b \equiv a^i \pmod{p}$ where $0 \leq i \leq (p-1)$

Algorithm :-

① Agree on Shared Public Parameters:

• Both parties agree on two vars

→ prime no. p ; a large prime no.

→ generator g ; a primitive root g modulo p , which is a no. that, when raised to different powers, produces a wide range of values modulo p .

② Private Key Generation

• Each party chooses a private key

→ Party A select a secret private key " a "

→ Party B " " " " b "

③ Public Key generation

- Using their private keys, both parties compute their public key.

$$\rightarrow \text{Party A computes } A = g^a \text{ mod } p$$

$$\rightarrow \text{Party B computes } B = g^b \text{ mod } p$$

- ④ Once both parties receive each other's public keys, they can compute a shared secret key using their private keys.

$\rightarrow$ Party A computes the shared secret as

$$S = B^a \text{ mod } p$$

$\rightarrow$ Party B computes the shared secret as

$$S = A^b \text{ mod } p$$

Ex: $g = 5, p = 23$

Alice selects $a = 6$ as private key

Bob selects $b = 15$ as private key

- Both values of S will be equal because of the properties of modular arithmetic.

$$\begin{aligned} S &= A^a \text{ mod } p = B^a \text{ mod } p \\ &= (g^b \text{ mod } p)^a \text{ mod } p \\ &= g^{ab} \text{ mod } p \\ &= (5)^{6 \times 15} \text{ mod } 23 \\ &= (5)^{90} \text{ mod } 23 \end{aligned}$$

Attacks

Man-in-the-Middle (MIM) attack :-

- Diffie-Hellman by itself does not authenticate the communicating parties, making it vulnerable to MIM attacks.

An attacker could intercept public keys and establish two separate shared keys, one with Party A and another with Party B, relaying communications between them while deciphering the msgs.

- Solution: To prevent MIM attacks, Diffie-Hellman is often combined with digital signatures or public key certificates to verify the identities of communicating parties.

Q3^{or} Demonstrate the process of encryption and decryption using El-Gamal encryption algorithm?

Ans. # El-Gamal encryption algorithm

- El-Gamal algo. is a public-key cryptosystem.
- It uses asymmetric key encryption for communicating b/w two parties and encrypting the msg.
- The cryptosystem is based on the difficulty of finding discrete logarithms in a cyclic gp that is even if we know g and g^k , it is extremely difficult to compute k .
- It was invented by Tahar ElGamal in 1985.

⇒ Process of working :-

① Key Generation

- Public parameters - select a large prime no. p and a generator g of the multiplicative gp $(\mathbb{Z}^*, p)$.
- Private Key: select a private key x such that $1 \leq x \leq p-2$.

• Public key: Compute $h = g^x \bmod p$. The public key is (p, g, h) and the private key is x .

② Encrypt

- To encrypt - a msg M
 - choose a random integer k such that $1 \leq k \leq p-2$
 - compute $c_1 = g^k \bmod p$
 - compute $c_2 = M \cdot h^k \bmod p$
 - The ciphertext is (c_1, c_2)

$$h^k = g^{kx}$$

③ Decrypt

- To decrypt - the ciphertext (c_1, c_2) using the private key x :
 - compute the shared secret $s = c_1^x \bmod p$
 - compute $s^{-1} \bmod p$
 - compute the original msg $M = c_2 \cdot s^{-1} \bmod p$

$$q \rightarrow p, A \rightarrow g, xA \rightarrow x, yA \rightarrow h, h^k = K$$



→ The Elgamal Cryptosystem

Global Public Elements

p
 g

prime no.
 $g < p$ and g is a primitive root of p

Key Generation by Alice

Select - private x

Calculate h

Public Key

Private Key

$$x < p-1$$

$$h = g^x \bmod p$$

$$\{p, g, h\}$$

Encryption by Bob with Alice's Public Key

Plaintext

Select - random integer k

Calculate S

Calculate C_1

Calculate C_2

Ciphertext

$$M < p$$

$$k < p$$

$$S = (h)^k \bmod p$$

$$C_1 = g^k \bmod p$$

$$C_2 = S \cdot M \bmod p$$

$$(C_1, C_2)$$

Decryptⁿ by Alice with Alice's
Private Key

Ciphertext

(C_1, C_2)

Calculate S

$$S = (C_2)^b \bmod p$$

Calculate Plaintext

$$M = (C_1 \cdot S^{-1}) \bmod p$$

Ex: $p=23, g=5, b=6$ (x)

• Bob computes

$$B = 5^6 \bmod 23 = 15625 \bmod 23 = 8$$

(^{private key}
~~public key~~)

• Alice wants to send an encrypted msg to Bob. $M=13$.
let's say

• let $k=10$ (random integer)

$$C_1 = 5^{10} \bmod 23 =$$

$$9765625 \bmod 23 = 9$$

$$C_2 = 13 \cdot 8^{10} \bmod 23 = 2$$

• Alice sends (C_1, C_2) to Bob

• To decrypt, Bob uses ^{his} private key $b=6$

$$S = 9^6 \bmod 23 = 531441 \bmod 23 = 9$$

• Bob then computes the modular inverse of 9 mod 23

$$m = 2 \cdot 18 \bmod 23 = 36 \bmod 23 = 13$$

• Thus Bob successfully decrypts the msg

$$\underline{\underline{m=13}}$$

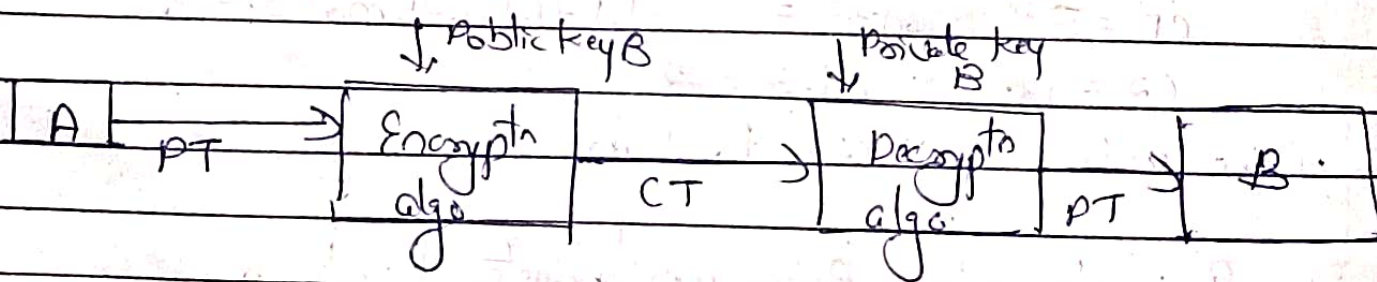
Limitation

- ① Slow processing - it is slower compared to other encryption algorithms, especially when used with long keys.
- ② Key size - require larger key size to achieve the same level of security as other algorithms; this can make it more difficult to use in some applications.

8m (Pro)

Q3 Explain the RSA with example & discuss its security?

Ans. RSA is the most common public-key algorithm, named after its inventors Rivest, Shamir, and Adelman (RSA).



- The idea of RSA is based on the fact that it is difficult to factorize a large integer.
- The public key consists of two no.s where one no. is a multiplicate of two large prime no.s.

- And private key is also derived from the same two prime no's.
- So, if somebody can factorize the large no., the private key is compromised.
- Therefore, encryption strength lies in the key size and if we double or triple key size, the strength of encryption increases exponentially.

⇒ RSA works :-

① Key Generation

- RSA key generation involves creating a public key for encryption and a private key for decryption.

- Choose two large prime no's 'p' and 'q'.
- Compute n : $n = p \times q$. The modulus 'n' is part of public key and determines the size of RSA key.
- Calculate Euler's Totient function: $\phi(n)$; Since $n = p \times q$, $\phi(n) = (p-1) \times (q-1)$.
- Choose a public exponent e : Select an integer e such that $1 < e < \phi(n)$ and 'e' is coprime to $\phi(n)$, $e = 3, 17, 65537$.

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(c) determine the private exponent d : calculate d as the modular inverse of e modulo $\phi(n)$, so
$$d \times e \equiv 1 \pmod{\phi(n)}$$

→ The public key is the pair (n, e) , and private key is (n, d) .

→ The p and q are discarded after this step.

(2) Encrypts: To encrypt msg M with PK

(a) Convert msg M into a integer m such that
$$0 \leq m < n$$

(b) Compute the ciphertext c using the public key (n, e)

$$c = m^e \pmod{n}$$

(c) Convert c back to original plaintext msg m

(3) Decrypts: To decrypt the ciphertext c with private key (n, d) :

(a) Recover the original msg m by computing

$$m = c^d \pmod{n}$$

(b) Convert m back to original plaintext msg M .

Example;

- Suppose $p=11$, $q=13$, giving us $n=143$ and $\phi(n)=120$
- We choose $e=7$, since $\gcd(7, 120)=1$.
Using extended Euclidean algorithm, we can compute $d=103$,
Since $7 \times 103 \equiv 1 \pmod{120}$
- Our public key is $(143, 7)$ and our private key is $(143, 103)$
- Suppose we want to encrypt the msg 42.
Using the public key, we compute the ciphertext
$$c = 42^7 \pmod{143} = 81$$
- To decrypt the CT, we use private key to compute $m' = 81^{103} \pmod{143} = 42$,
which is original msg

Security foundation of RSA :-

① Difficulty of factoring large integers :-

The security of RSA depends on the difficulty of factoring the large modulus n , which is product of two large primes no's. For sufficiently large ' n ', factoring computationally is infeasible.

② Difficulty of computing the modular inverse:

The decrypt key 'd' is derived by computing the modular inverse of e modulo $\phi(N)$.

This is possible only if the values of p & q are known.

③ Assumptions of Prime Selection:

RSA's security also depends on using primes that are chosen carefully to avoid common factoring techniques.

⇒ read common attacks in RSA and timing attack from PDF

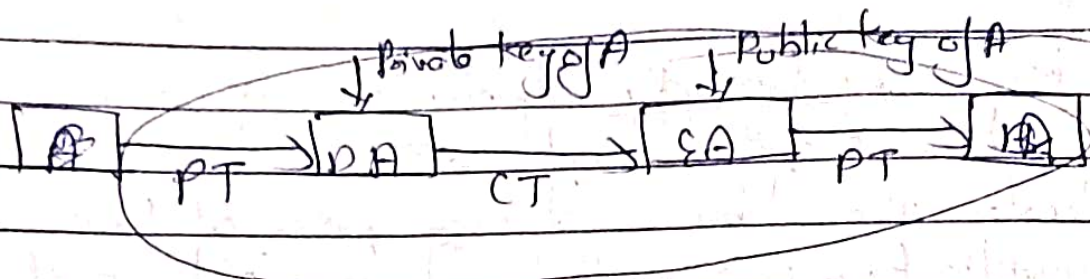
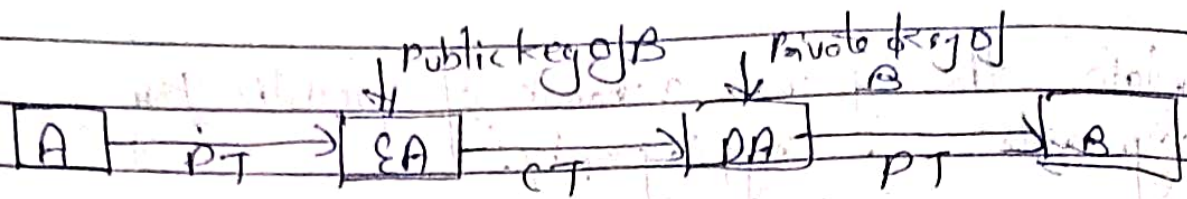
Q4 Explain the general scheme of public key asymmetric cryptography.

Ans.

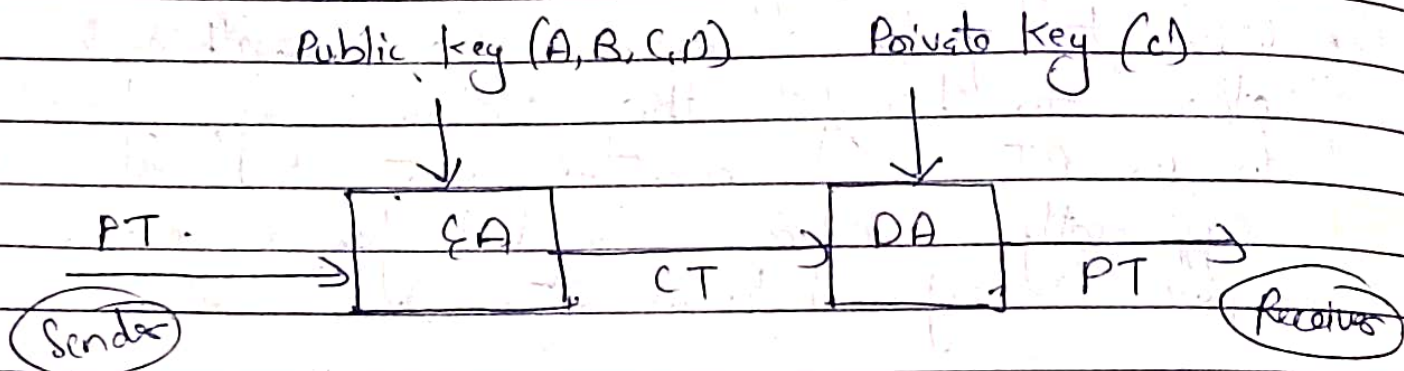
- Public Key cryptography provides a secure way to exchange information and authenticate users by using pairs of keys.
- The public key is used for encryption and signature verification, while the private key is used for decryption and signing.
- When two parties communicate with each other to transfer the sensible msg, referred to as PT, is converted into apparently random unreadable for security purposes referred to as CT.

⇒ Public Key encryption

- Public keys of every user are present in the Public Key Register.
- If B wants to send a confidential msg to C, then B encrypts the msg using C's Public Key.
- When C receives the msg, then C can decrypt it using its own private key.
- No other recipient other than C can decrypt the msg because only C knows its private key.



Public Key Encryption



Components :-

- ① Plain text - This msg is readable and understandable
- ② Cipher text - The (CT) is produced as an output of encryption algo.
- ③ $\{A\}$ - convert PT to CT
- ④ $\{B\}$ - convert CT to PT
- ⑤ Public & Private key - one key either Private key or public key is used for encryption and other for decryption

→ Weakness

- ① Public key encrypts is vulnerable to Man-in-the-middle attack
- ② This algo. also fails when the user lost his private key.
- ③ Public key is also weak towards MITM attack.

→ Application

- Encrypts / Decrypts →
- ① Digital signature
 - ② Key exchange

Private key encrypts (Symm)
One key for both encrypts and decrypts

Relies on secret key management

Fast, suitable for large data

Requires secure sharing of secret key

Algo; AES, DES, Triple DES

Public key encrypts (asymm)
① Public key for encrypts
private key for decrypts.

② Solves key distribution due to public-private encryption

③ Slower, best for small data or key exchange

④ Easier with public key openly shared

⑤ Algo; RSA, Diffie-Hellman.